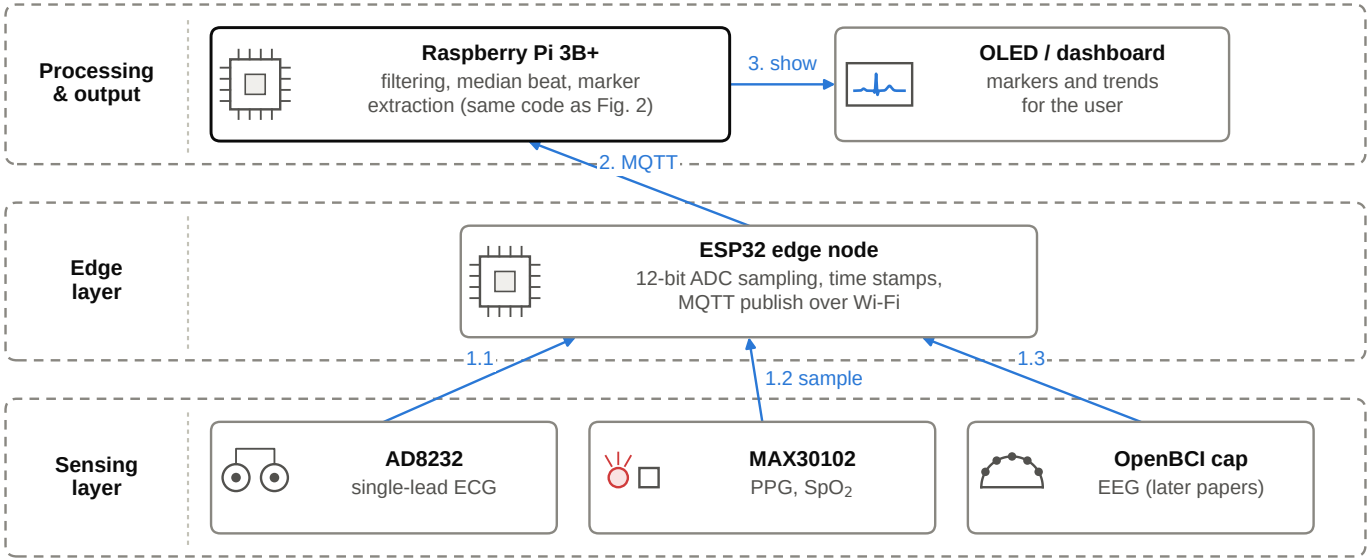
Processing & output
Raspberry Pi 3B+
filtering, median beat, marker extraction (same code as Fig. 2)
3. show
OLED / dashboard
markers and trends for the user
2. MQTT
Edge layer
ESP32 edge node
12-bit ADC sampling, time stamps, MQTT publish over Wi-Fi
1.1
1.2 sample
1.3
Sensing layer
AD8232
single-lead ECG
MAX30102
PPG, SpO2
OpenBCI cap
EEG (later papers)